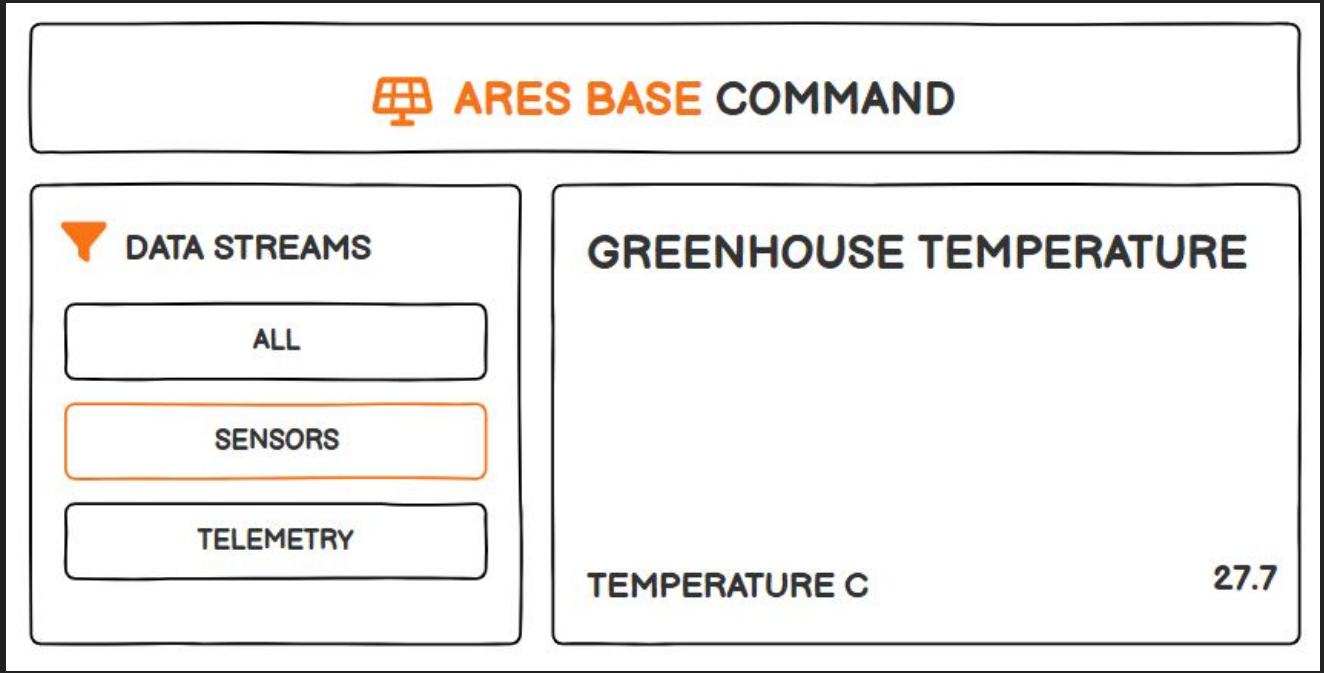
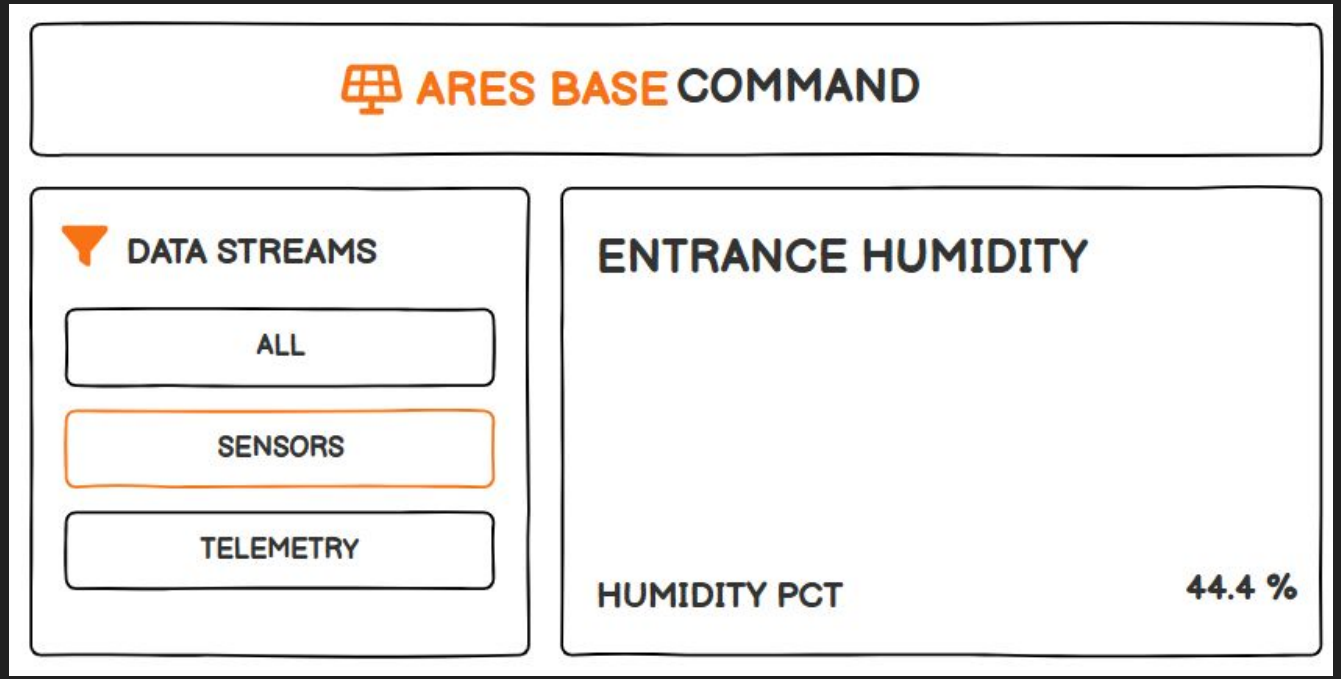
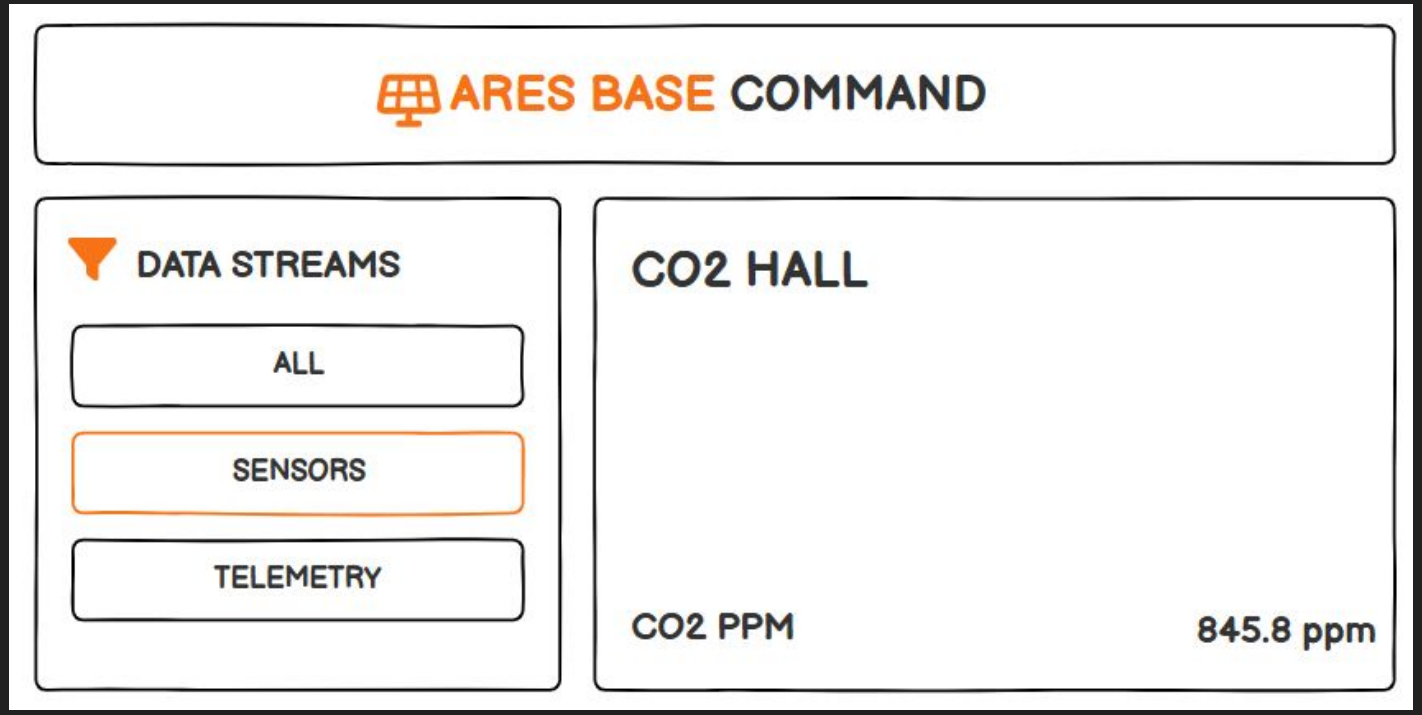
As a Habitat Operator, I want to view the real-time value of the greenhouse temperature sensor so that I can ensure the plants do not freeze.
(NFR: Fast UI update based on the in-memory cache)



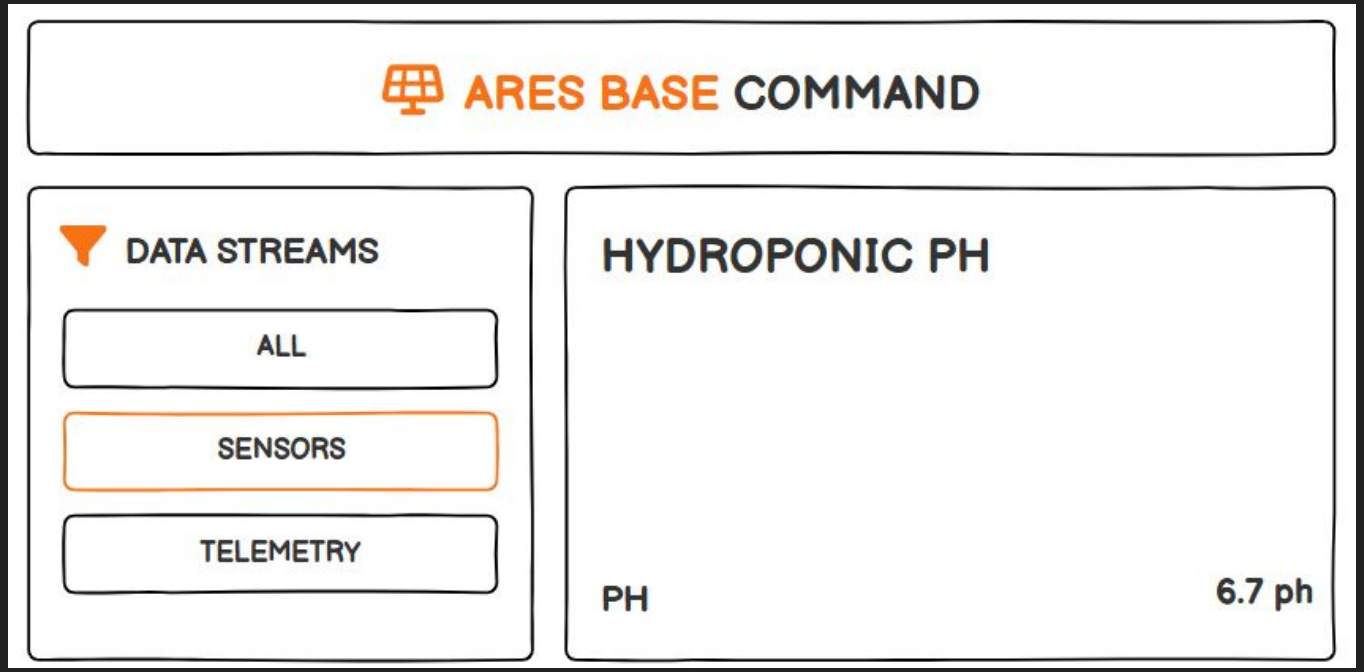
As a Habitat Operator, I want to monitor the entrance_humidity so that optimal vital parameters are maintained.



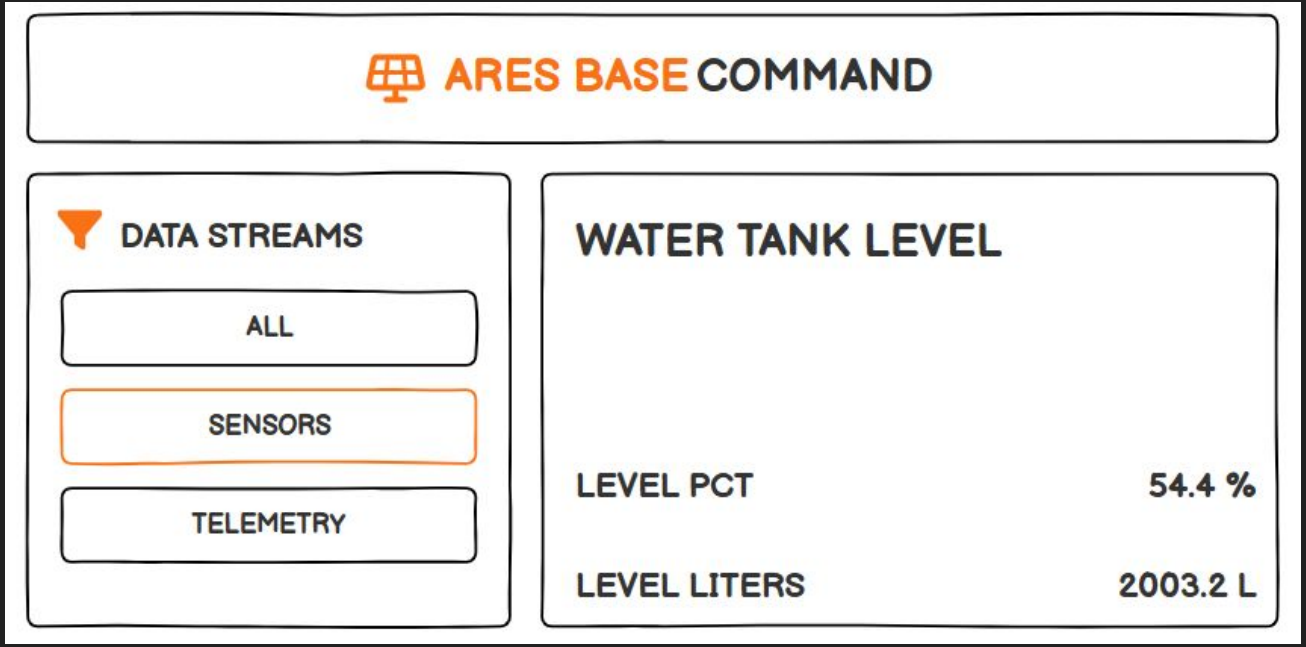
As a Habitat Operator, I want to see the co2_hall levels so that the air remains breathable.



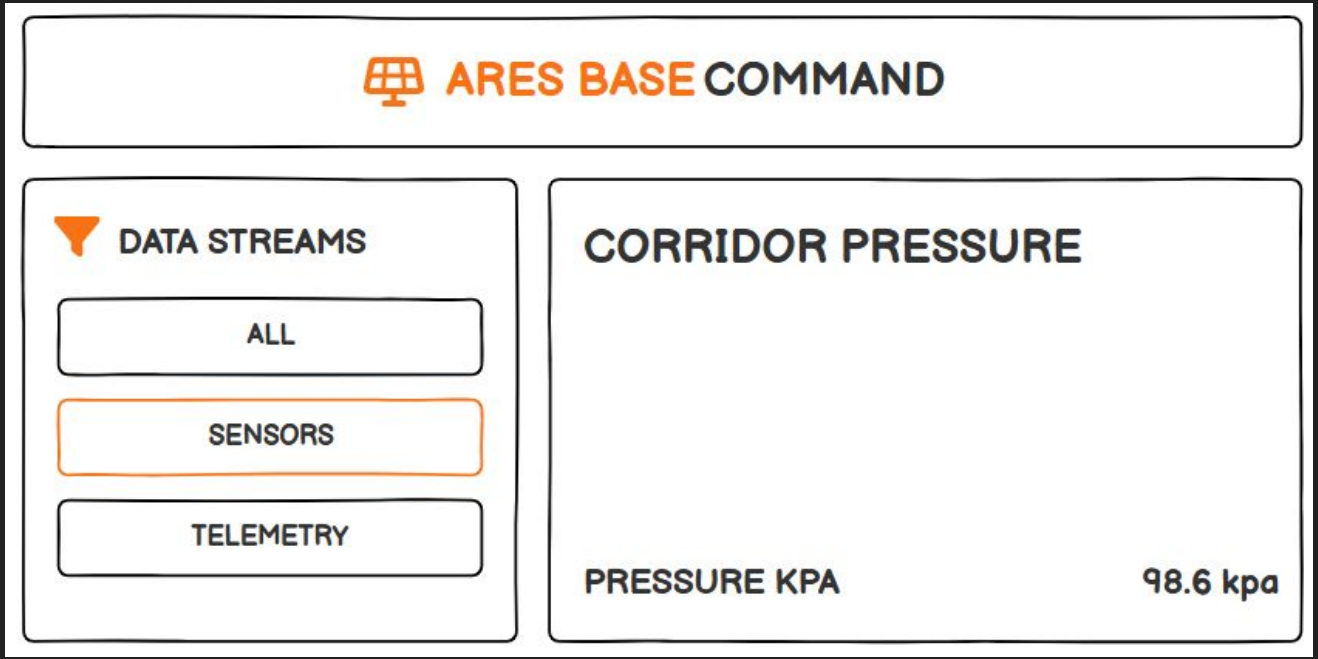
As a Habitat Operator, I want to check the hydroponic_ph so that nutrients are correctly absorbed by the crops.



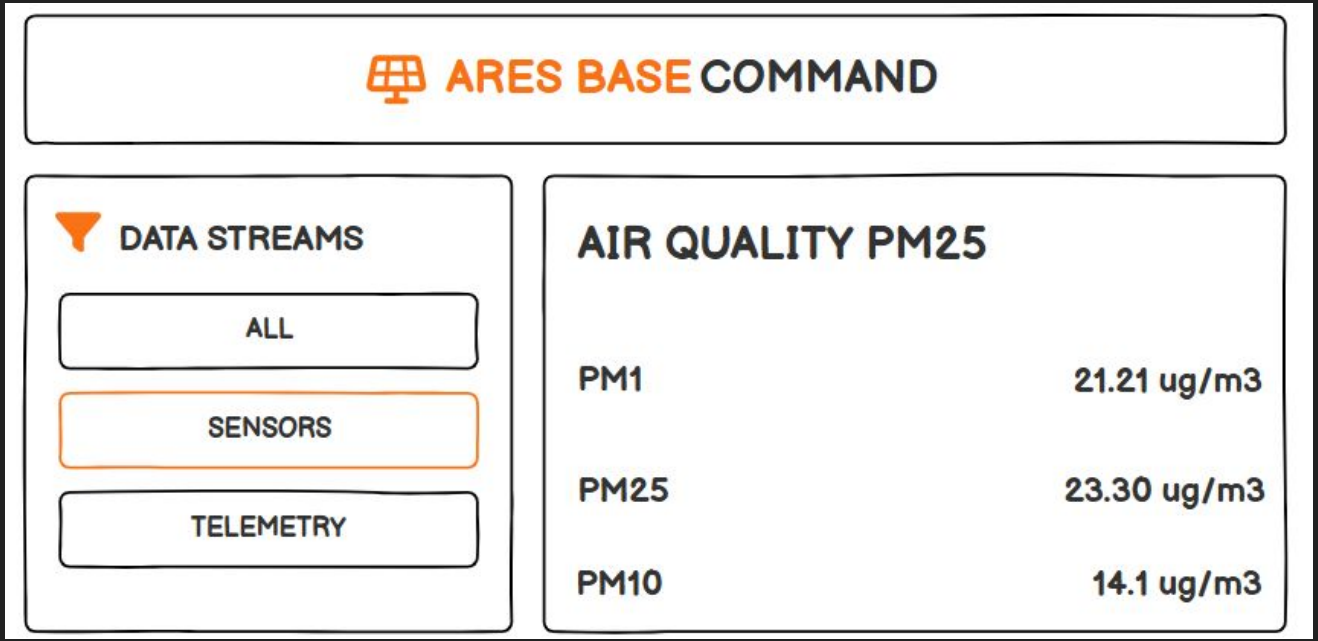
As a Habitat Operator, I want to view the water_tank_level so that the habitat does not run out of water resources.



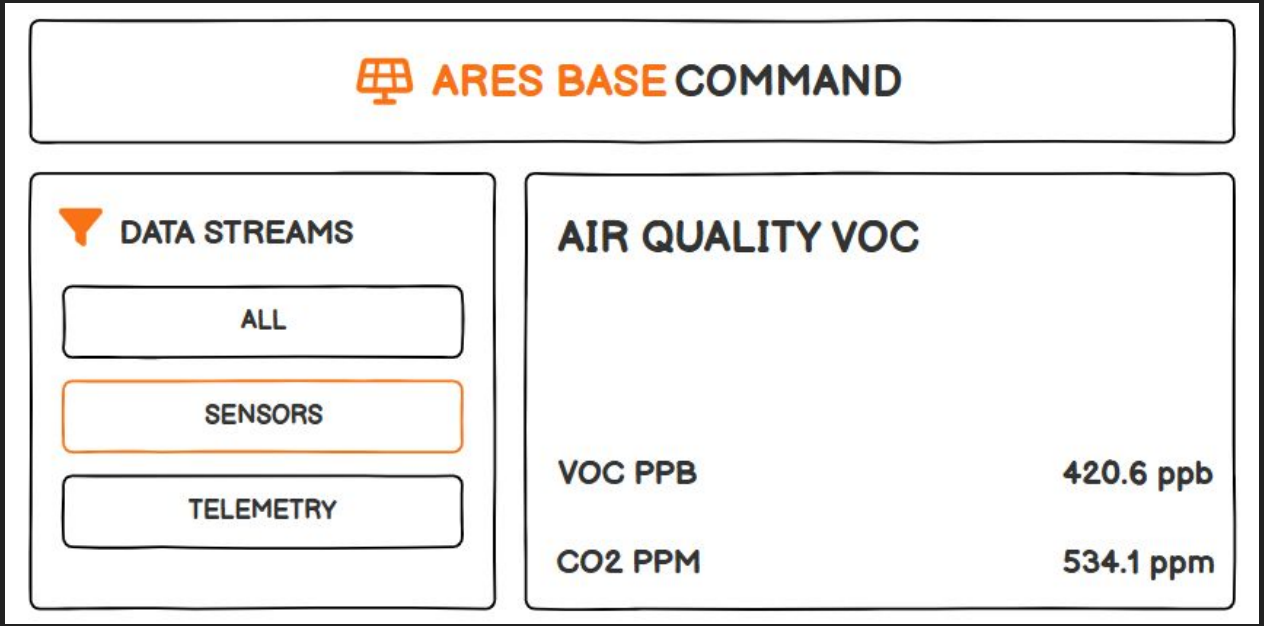
As a Habitat Operator, I want to monitor the corridor_pressure so that fatal depressurizations are prevented.



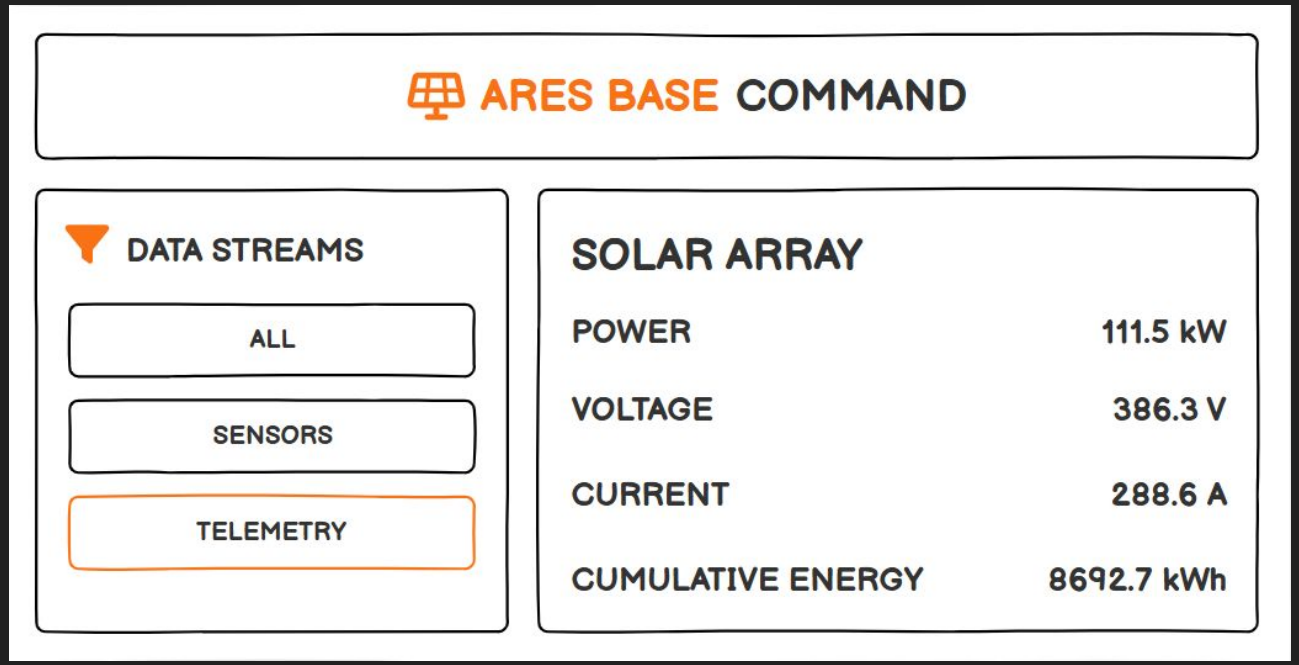
As a Habitat Operator, I want to see the PM 1, 2.5, and 10 levels from air_quality_pm25 so that I can assess the breathable air quality.



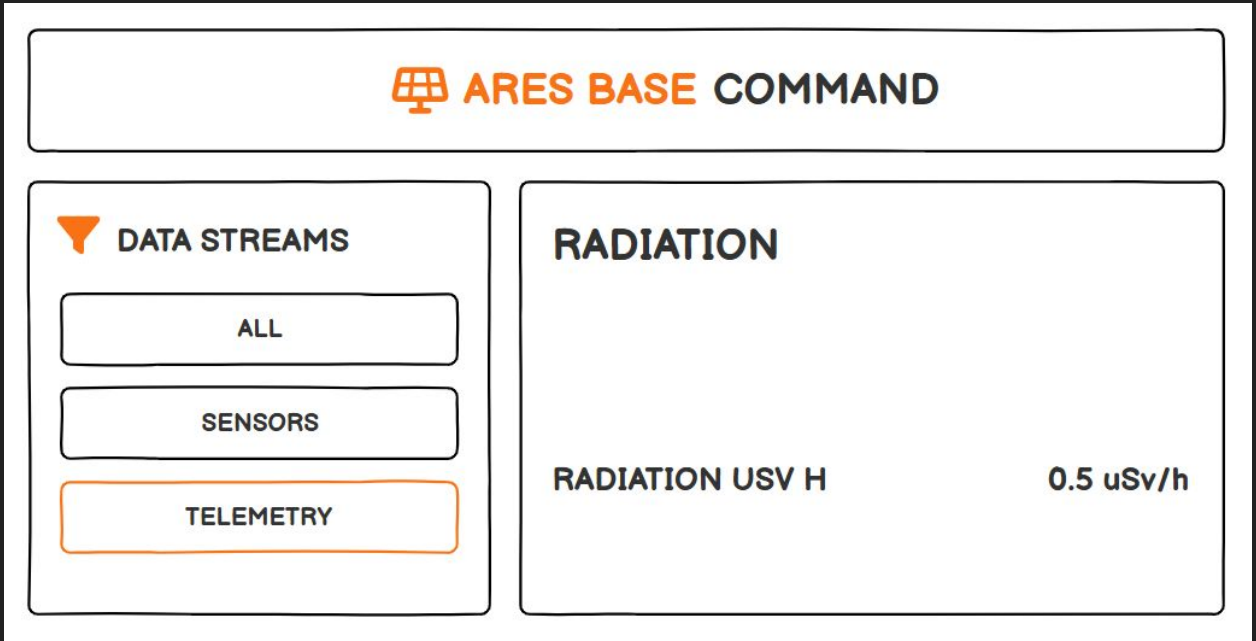
As a Habitat Operator, I want to monitor the volatile organic compounds (air_quality_voc) so that intoxications are prevented.



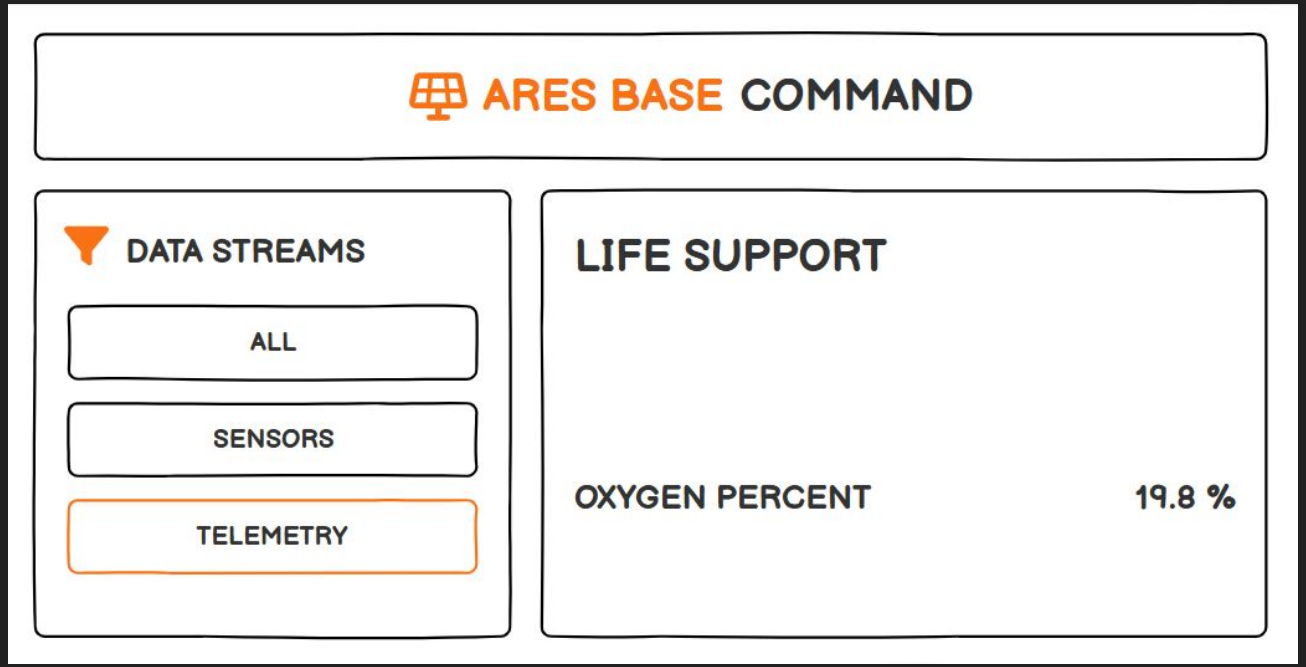
As a Habitat Operator, I want to view the power generated by the mars/telemetry/solar_array so that I can ensure the base has sufficient energy.



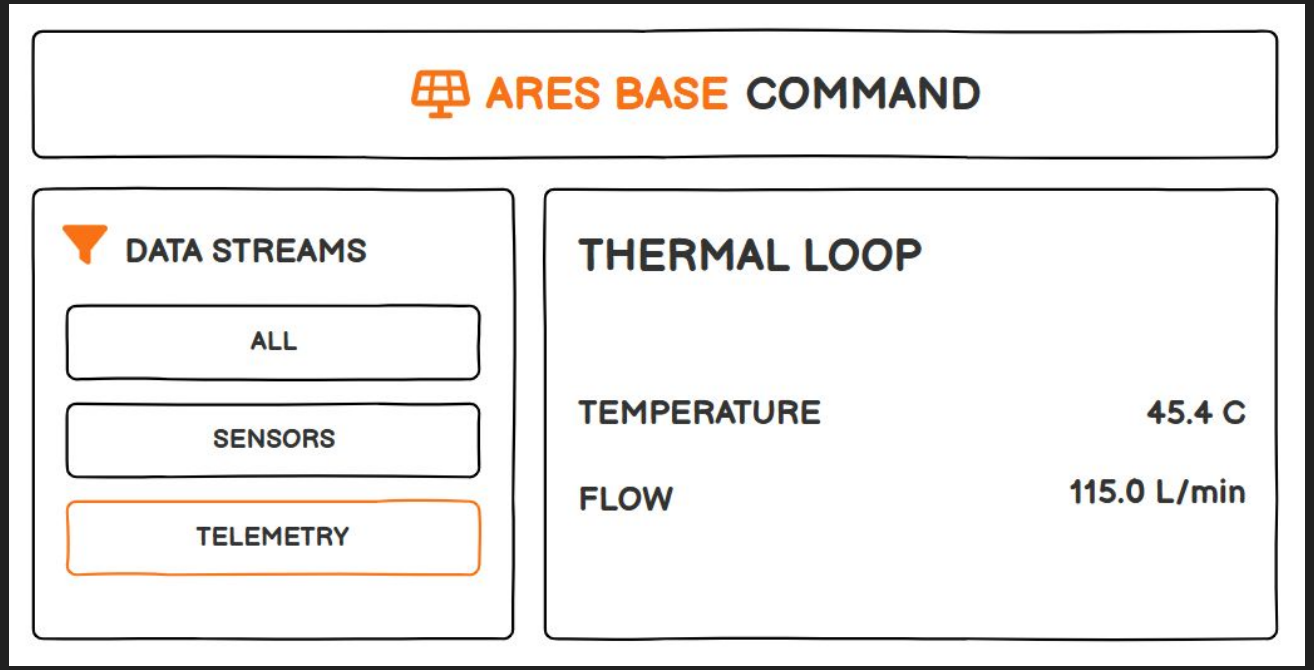
As a Habitat Operator, I want to monitor external mars/telemetry/radiation levels so that astronauts do not exit the habitat during solar storms.



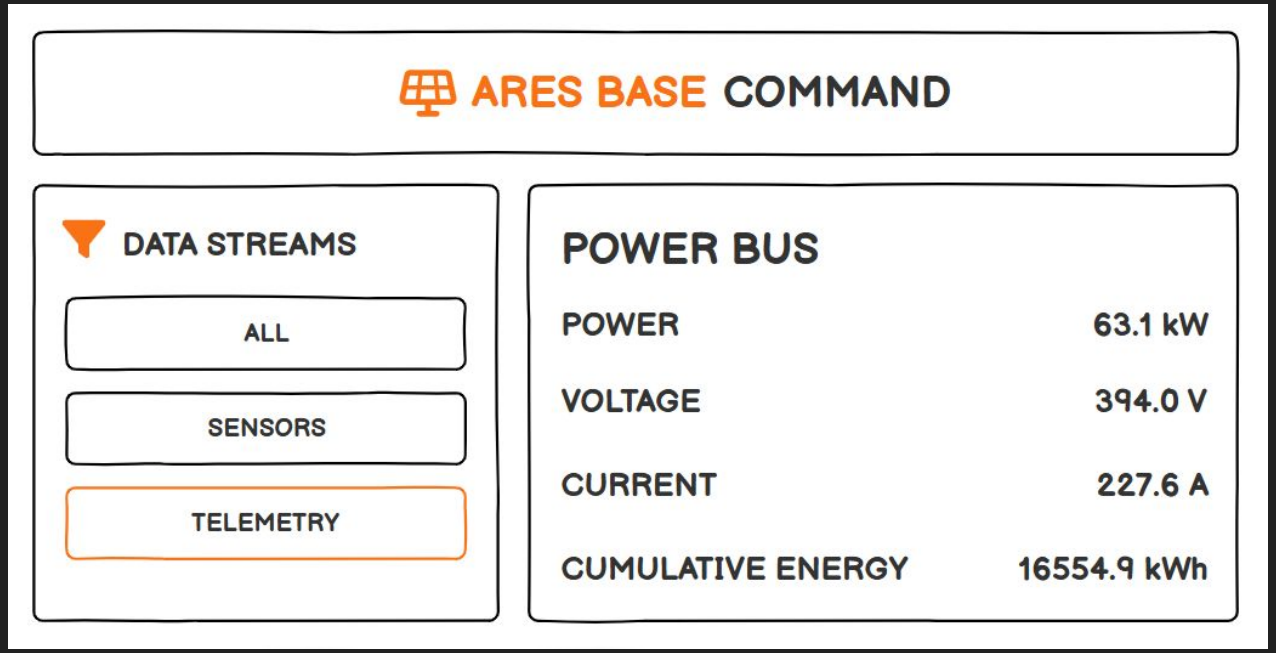
As a Habitat Operator, I want to view the real-time status of mars/telemetry/ life_support so that critical systems are verified as operational.



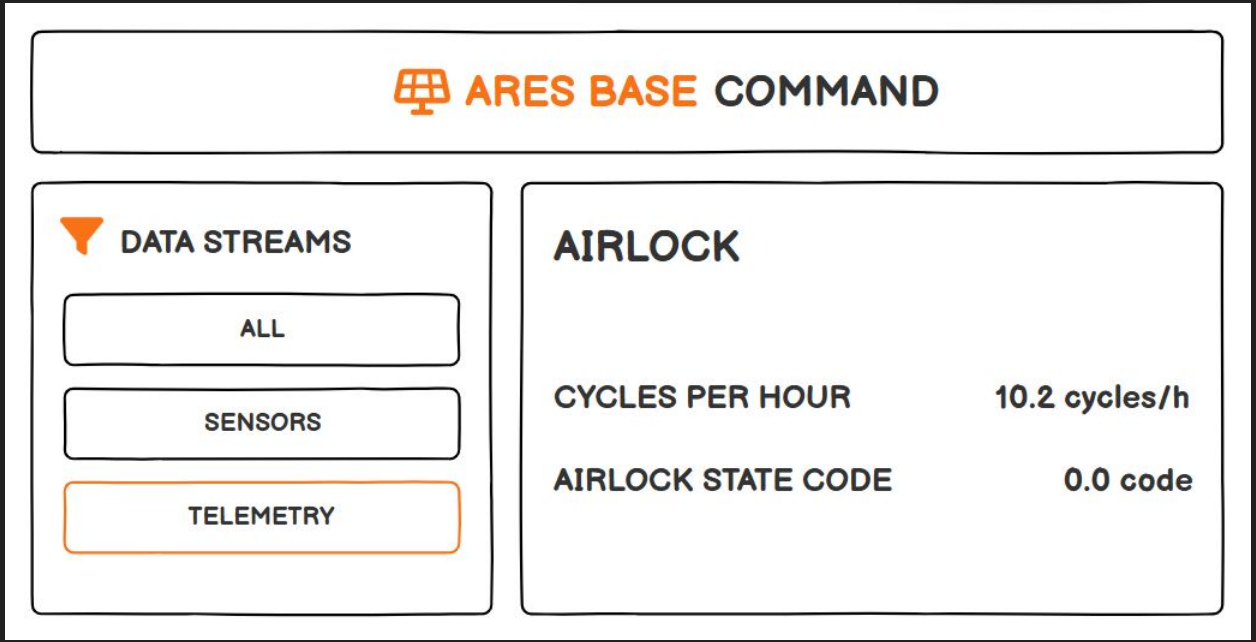
As a Habitat Operator, I want to monitor the temperature and flow of the mars/telemetry/thermal_loop so that the habitat does not overheat.



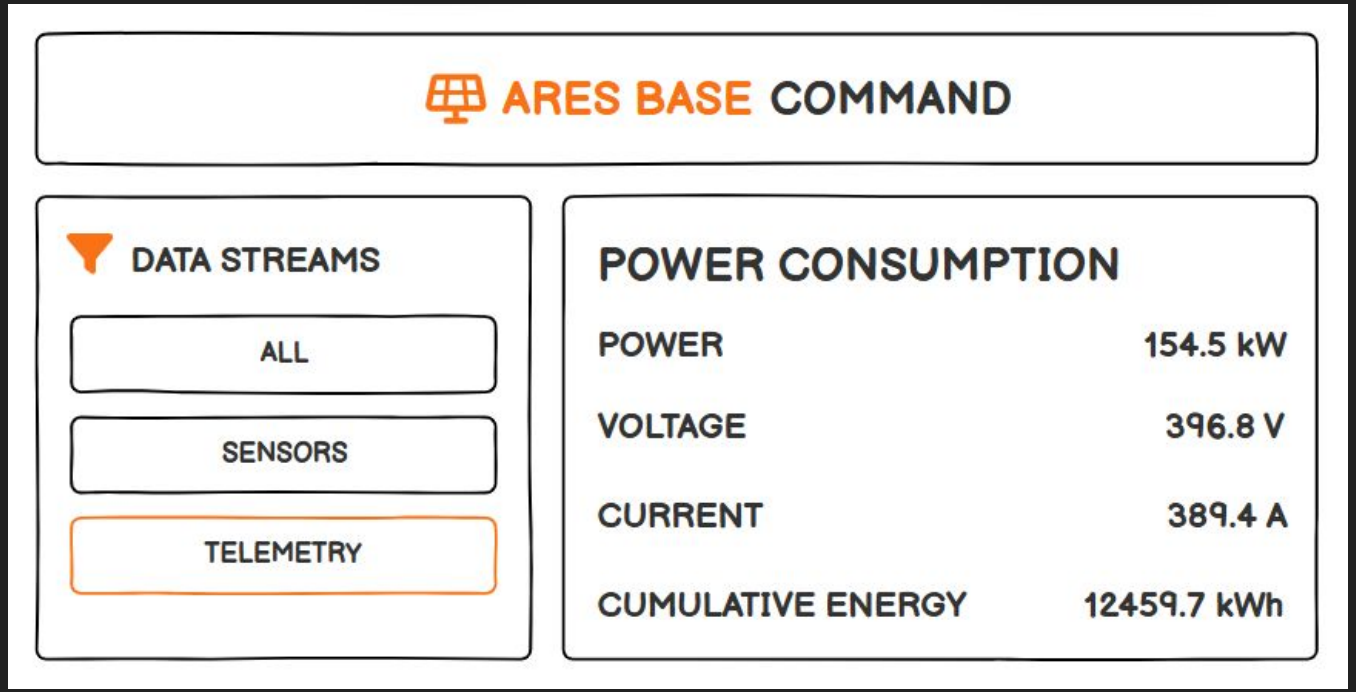
As a Habitat Operator, I want to monitor the voltage and current of the mars/telemetry/power_bus so that short circuits are prevented.



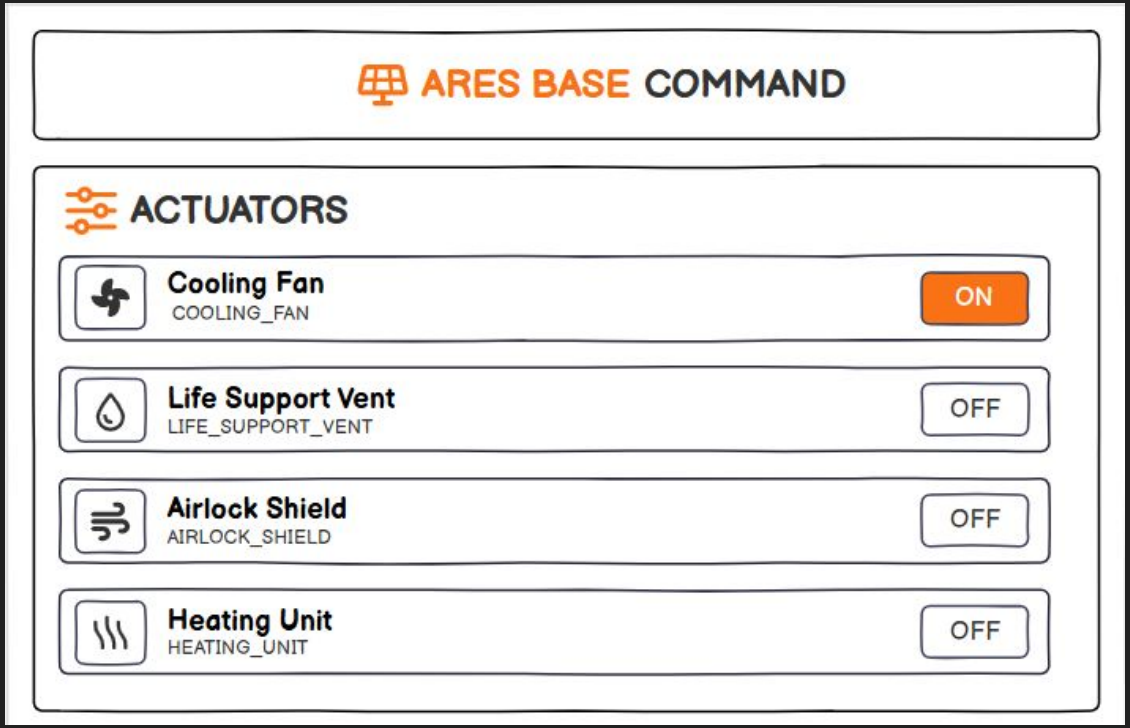
As a Habitat Operator, I want to view the status of the mars/telemetry/airlock so that I know when the doors are pressurizing or depressurizing.



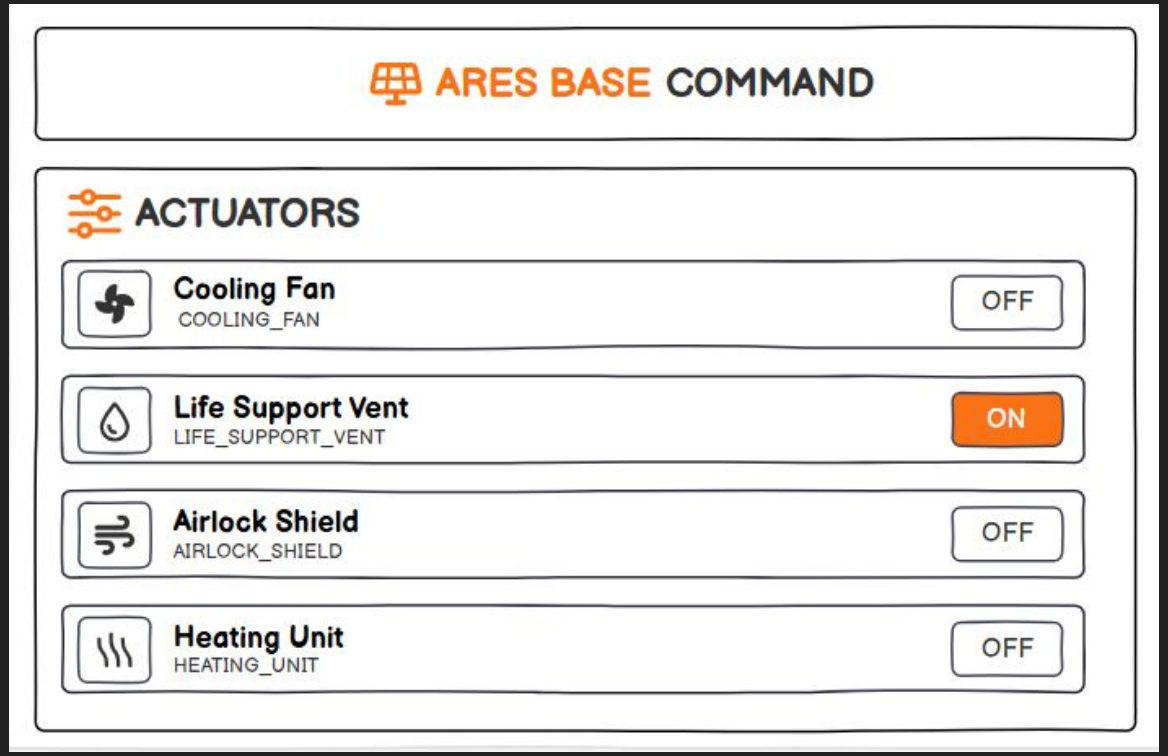
As a Habitat Operator, I want to see the global mars/telemetry/power_consumption so that energy rationing can be planned if necessary.



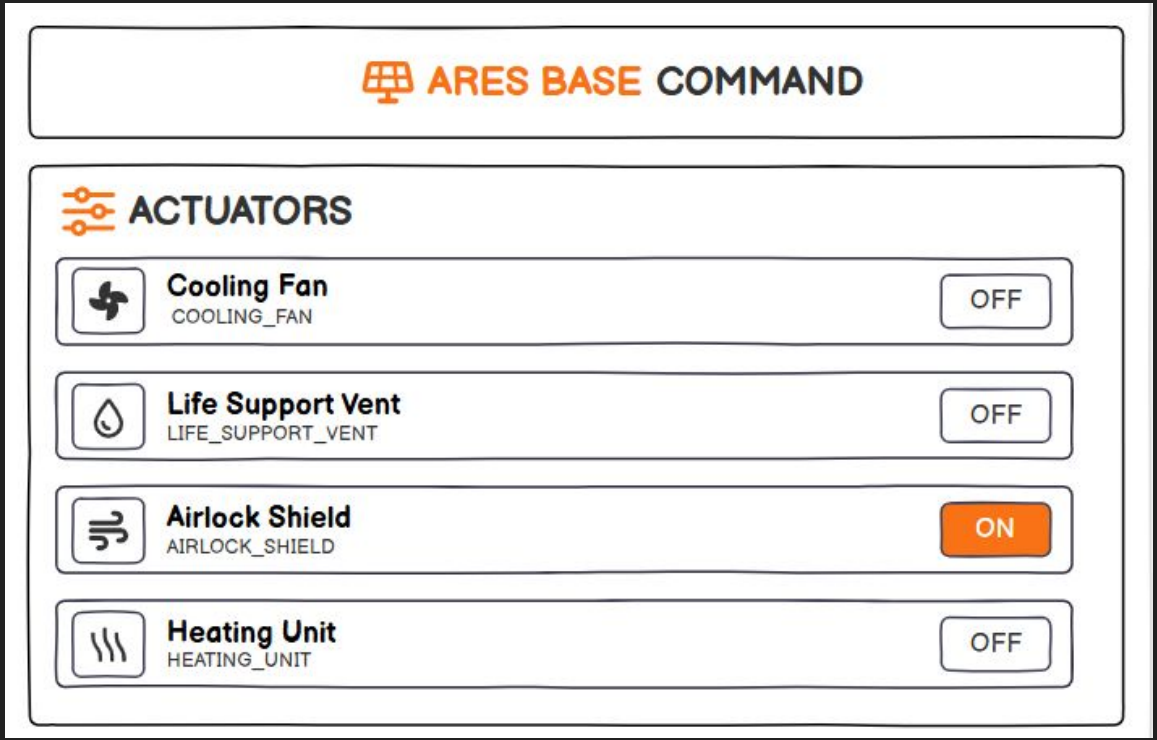
As a Habitat Operator, I want a dashboard button to manually toggle the cooling_fan so that I can force a temperature drop.
(NFR: The UI must update the actuator state via REST POST request)



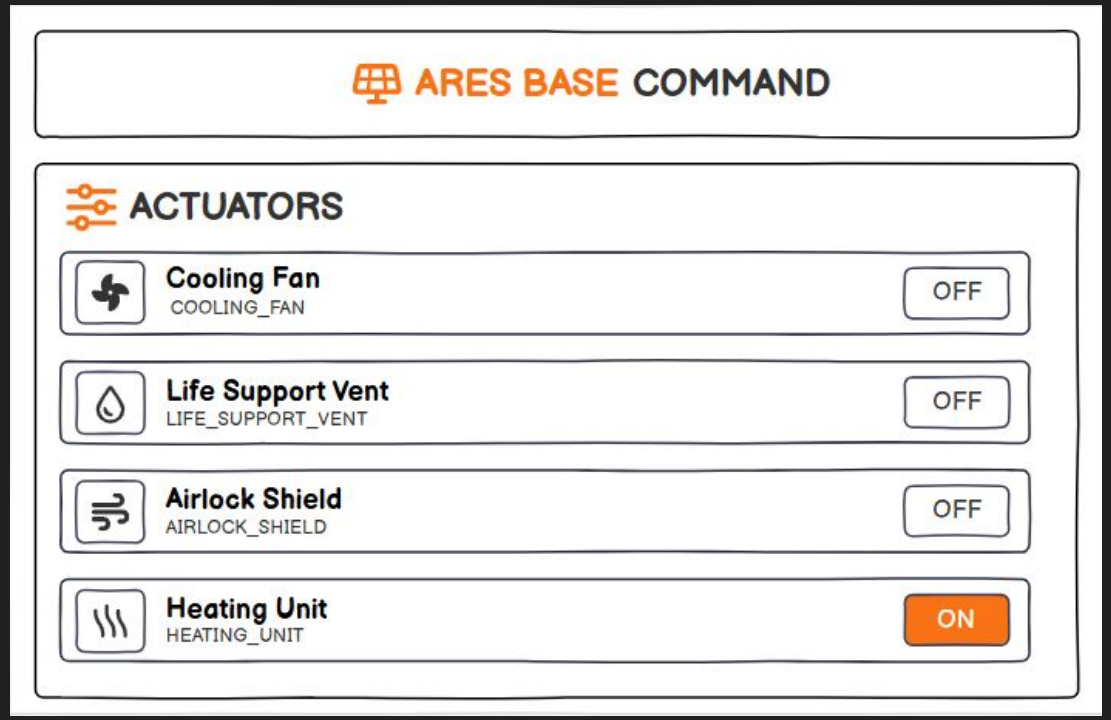
As a Habitat Operator, I want to be able to toggle the life_support_vent so that I can manually correct oxygen levels.



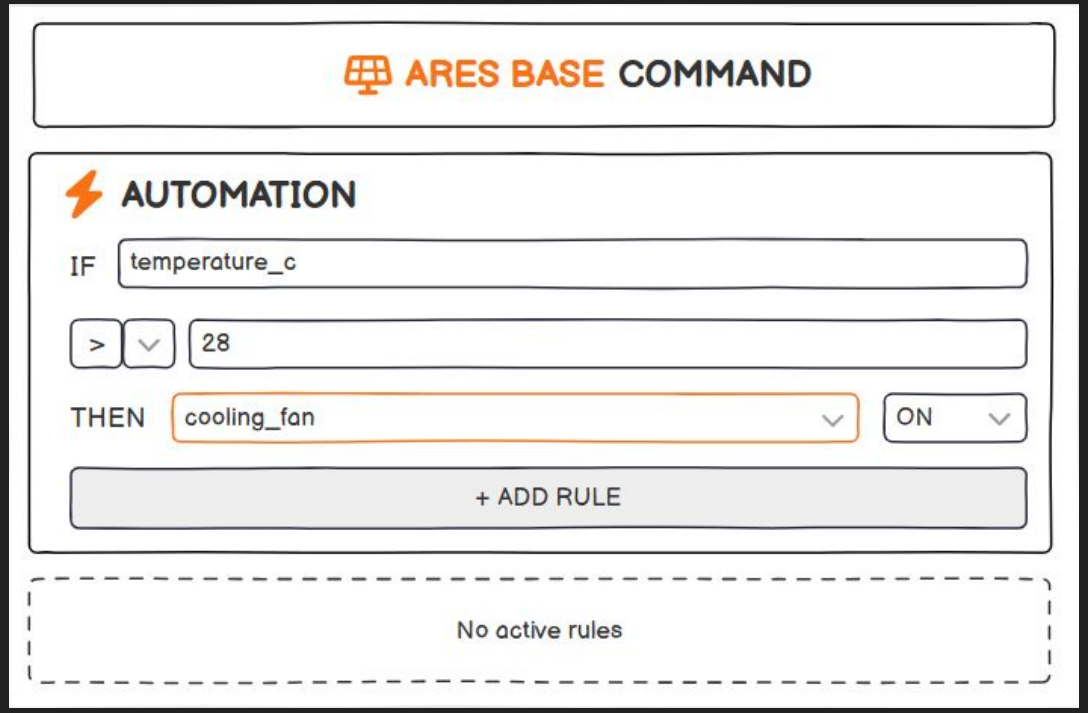
As a Habitat Operator, I want to be able to toggle the airlock_shield so that I can manually protect the base from radiation.



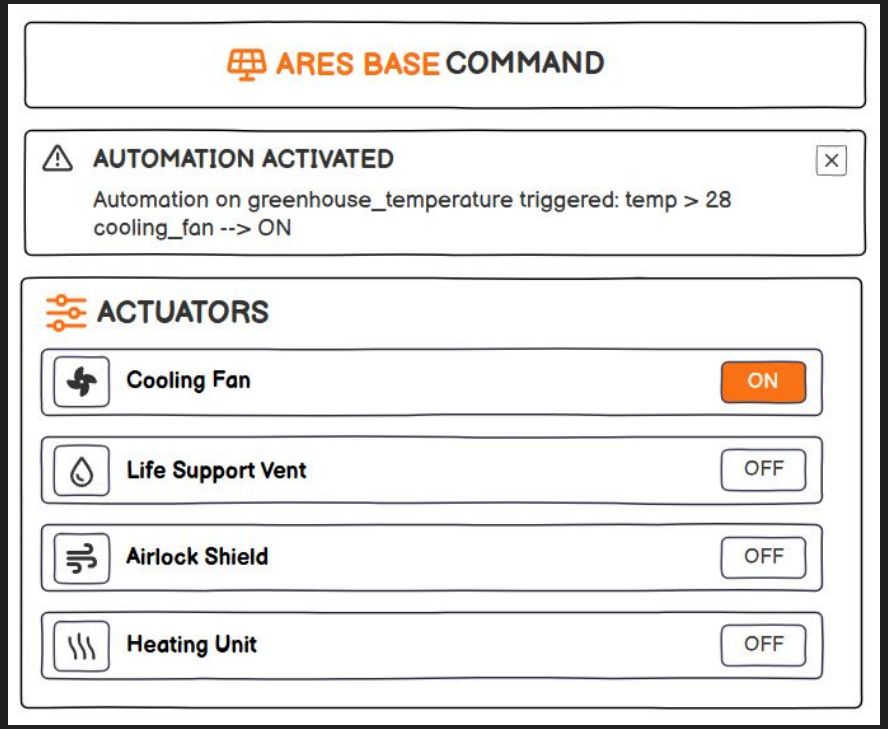
As a Habitat Operator, I want to toggle the heating_unit so that the temperature does not drop below the survival threshold.



As a Habitat Operator, I want a front-end interface to create a new IF-THEN rule so that the system can react autonomously to environmental changes.
(NFR: Rules must be saved in a persistent database)

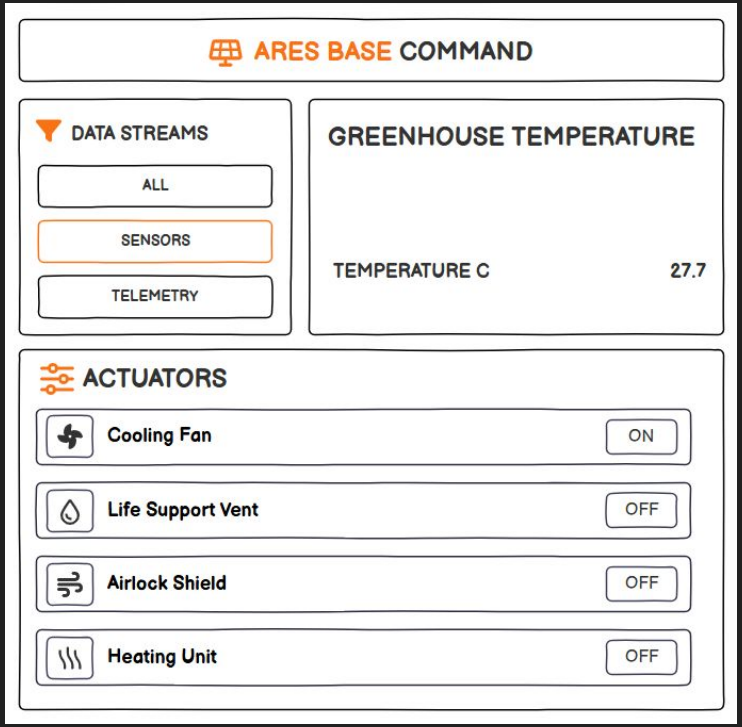


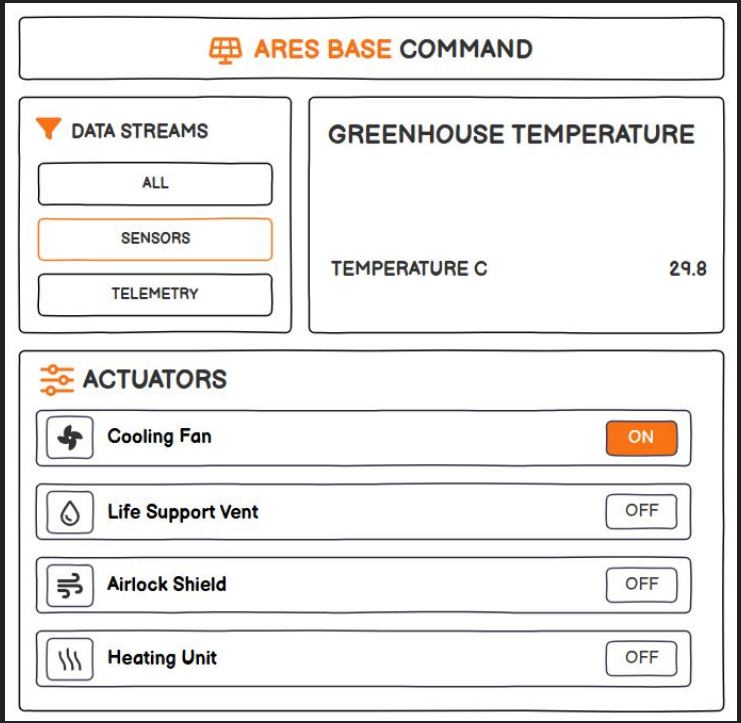
As a Habitat Operator, I want the automation engine to immediately execute the target action (e.g., turn an actuator ON or OFF) whenever a rule's condition is met, so that the habitat can react autonomously.



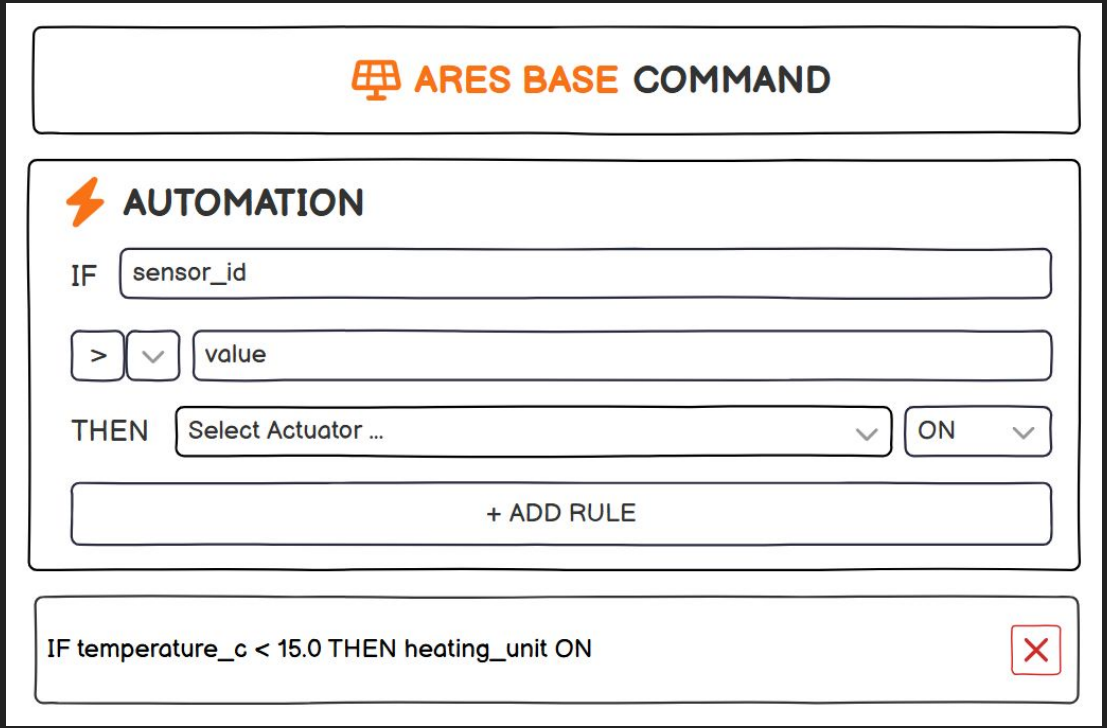
As a Habitat Operator, I want the dashboard to automatically update the displayed state of an actuator whenever it is modified by an automation rule, so that I am always aware of the system's current status in real-time.

NFR: The UI must update via WebSocket/Socket.io without refreshing

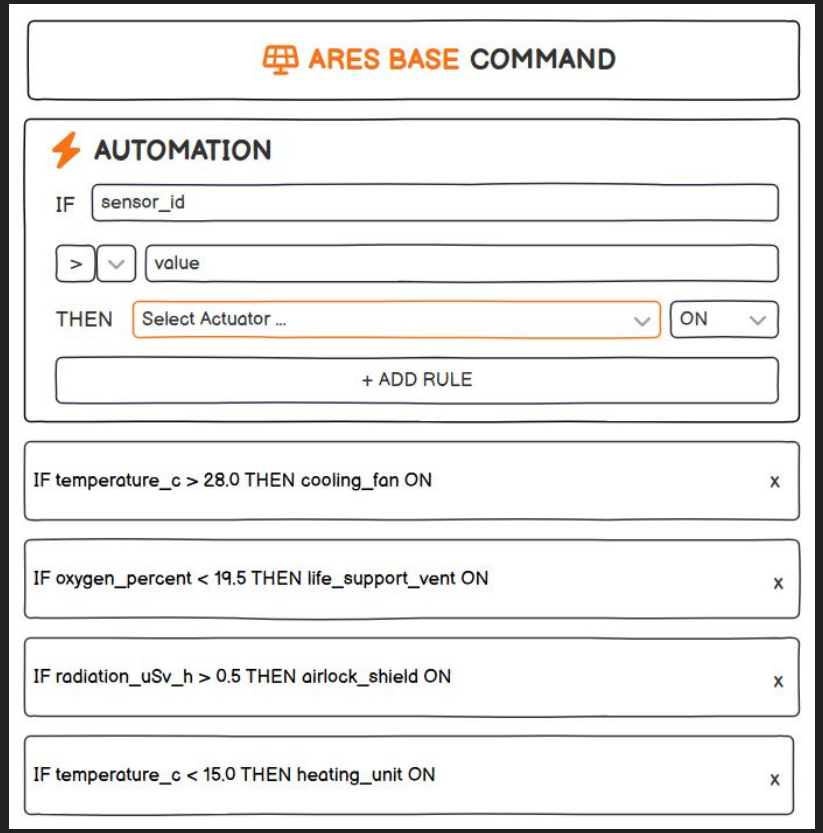
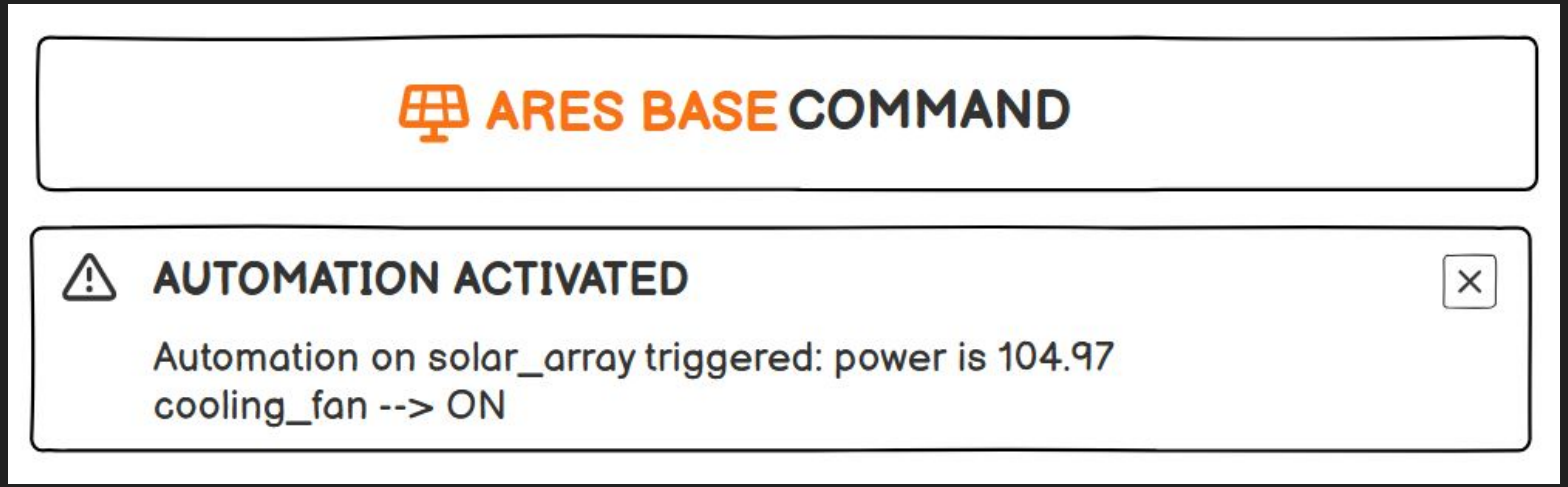




As a Habitat Operator, I want to be able to delete an existing rule so that I can remove obsolete or incorrect automations.



As a Habitat Operator, I want to view the list of all active automation rules via the frontend so that I have an overview of the automatic controls in place.



As a Habitat Operator, I want to receive real-time alert notifications (pop-up) on the dashboard, so that I am aware whenever an automation rule is triggered.